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Strategic Network Response Brief

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Summary of Network-Level Activity

After reviewing the Snort_alert_logs, Zeek_dns_logs, and Zeek_conn_logs, I found multiple indicators of possible malicious activity that could be a malware infection turning into a command-and-control (C2) communications attack with possible ransomware behavior. I found that host 192.168.1.101 queried *malicious-domain.com*, which became 203.0.113.1, and then established a successful HTTP service connection to that same IP address according to the zeek_conn_log. This was an interesting find because the suspicious DNS request was immediately followed by a network communication with that host. And the amount of data being transferred could indicate a malware download. Also there is a SNORT alert labeled "ET POLICY Possible Powershell Command," which suggests that PowerShell could've been used to connect to an external host or download content; this technique is common with malicious activity and classified as a Command and Control tactic according to the MITRE ATT&CK framework.

Additional evidence I found increases my suspicion of the likelihood of a compromise. A snort alert labeled "ET TROJAN External C2 via HTTP POST" indicated a possible command-and-control traffic between 192.168.1.101 and 10.0.0.5, while the Zeek_conn_log confirmed a successful HTTP session between both systems. I also reviewed the "ET MALWARE Ransomware File Extension Activity" alert involving 192.168.1.102 and 192.0.2.50, with a corresponding connection activity recorded in the zeek_conn_log. The ransomware alert is a cause for concern because it could mean a file encryption or modification and disrupt operations. Based on the evidence collected, I believe the activity is a high possibility of it being malicious and needs immediate investigation and containment.

MITRE Mapping

Found Behavior & Logs	Suspected MITRE Tactic	Suspected Technique	Business Impact	Recommended Action

DNS request to malicious-domain.com	Command and Control	T1105 - Ingress Tool Transfer	**High risk**. Communication with a suspicious external host can indicate a malware delivery, unauthorized downloads, or attacker controlling communications. This could result in a system compromise.	Investigate DNS, firewall, and web proxy logs. Block the malicious domain and IP address after it's confirmed malicious.
SNORT Alert: ET POLICY Possible Powershell Command	Stealth	Sub-Technique: T1027.010 - Obfuscated Files or Information, Command Obfuscation	**High risk**. Attackers using PowerShell commands can be used to execute malicious code, download malware, and evade detection through command obfuscation.	Enable PowerShell logging on all systems, install Antivirus/Antimalware, and implement Attack Surface Reduction rules to restrict suspicious PowerShell activity.
Snort Alert: ET TROJAN External C2 via HTTP POST	Exfiltration	T1041 - Exfiltration Over C2 Channel	**High risk**. A compromised system could be sending sensitive information to an attacker controlling the infrastructure through an established command-and-control channel. This can result in a sensitive data breach or even further compromise of	Investigate HTTP POST traffic, outbound communications for suspicious data transfers, isolate the affected host, block malicious IP addresses, and investigate potential data loss or any unauthorized access.

			the business's assets.	
SNORT Alert: ET MALWARE Ransomware File Extension Activity	Impact	T1486 - Data Encrypted for Impact	**Very high risk**. Ransomware can encrypt important data, disrupt operations, and create significant recovery costs and downtime for businesses.	Immediately isolate affected systems, preserve forensic evidence, find the ransomware process, and restore data from backups after containment.

Business Impact

Based on my analysis of the SNORT alerts and Zeek logs, I believe the activity is malicious. Host 192.168.1.101 first resolved malicious-domain.com to 203.0.113.1 and then established a successful HTTP connection to that address. The connection transferred about 900 bytes outbound and 4,096 bytes inbound; which can mean that the system could've downloaded a payload or received instructions from an external host. This activity is further supported by the Snort alert "ET POLICY Possible Powershell Command," which indicates PowerShell could have been used to execute commands or retrieve malicious content. Additionally, a second alert "ET TROJAN External C2 via HTTP POST" found communication between 192.168.1.101 and 10.0.0.5, while zeek_conn_log recorded a successful HTTP session lasting 10.5 seconds. The use of HTTP POST traffic connected with a C2 alert increases the possibility that data was being transmitted between a compromised host and attacker's external system.

The overall potential business impact is massive because the logs also revealed "ET MALWARE Ransomware File Extension Activity" involving 192.168.1.102 and 192.0.2.50, with a corresponding HTTP connection lasting 25 seconds. This activity suggests that files could've been encrypted or modified as part of a ransomware attack. It can create the risk of data loss and operation disruption. If the affected systems belong to regular employees, the impact can be limited to just local files and department resources. But if the compromised hosts belong to higher-level employees like administrators or users with access to sensitive business information, attackers could potentially gain access to financial records, sensitive data, or administrative credentials. The combination of suspicious DNS activity, inbound data transfers, PowerShell execution, command and

control communications, and ransomware activity requires immediate containment and investigation due to all evidence pointing to a high-risk incident.

Response Strategy Recommendation

From the evidence gathered from the Snort alerts and Zeek logs, I would recommend immediately isolating hosts 192.168.1.101 and 192.168.1.102 to prevent further communication with potentially malicious systems. The DNS query to malicious-domain.com, the following connection to 203.0.113.1, the transfer of 900 bytes outbound and 4,096 bytes inbound, and the PowerShell alert all suggests that 192.168.1.101 could've downloaded or executed malicious commands. The "ET TROJAN External C2 via HTTP POST" alert indicated possible command-and-control communications, which could have been used to exfiltrate data or receive attacker instructions. I would also block the domain and IP addresses, check PowerShell, firewall, and endpoint logs for other possible indicators of compromise. There will also be additional investigation into the 10.5 seconds HTTP session between 192.168.1.101 and 10.0.0.5. Since 192.168.1.102 also triggered a ransomware file extension alert and connected with 192.0.2.50 for about 25 seconds, I would preserve the forensic evidence, scan the endpoint, and identify any encrypted or modified files. This recommended response strategy will require time and resources; the benefits of preventing ransomware spread, protecting confidential data, and reducing any recovery costs will heavily outweigh the operational impact of the response measures.